

Design and Economic Analysis of a Stand-alone microgrid system using Dandelion Optimizer - A rural case in Southwest Morocco

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Abstract—The purpose of this paper is to present an optimization model designed to identify the ideal size for a standalone hybrid renewable energy system (HRES) that can effectively meet the energy requirements of a rural community located in Southwest Morocco while maximizing profitability. The HRES, which is composed of Photovoltaic (PV) panels, Wind Turbines (WTs), Hydrogen Tanks (HTs), Electrolyzer units, and a Fuel Cell (FC) system, will be designed to maximize profitability. The efficiency of DO is assessed in this study and is compared to commonly used optimization methods such as Genetic Algorithm (GA) and Particle Swarm Optimization (PSO). The findings reveal that DO has better convergence properties than GA and PSO and can generate optimal solutions in a relatively short amount of time. Our research provides a more sustainable and eco-friendly system for electrification and can serve as a valuable reference for investment in areas of Southwest Morocco where electricity is scarce.

Index Terms—Dandelion Optimizer, Renewable Energy, Microgrid, Optimal Design, Meta-heuristic Algorithm, Sizing

I. INTRODUCTION

In recent years, there has been a surge of interest in renewable energy sources as a viable solution to tackle the challenges brought about by the depletion of fossil fuels, mitigate the impacts of climate change, and meet the increasing demand for energy [1]. The scientific community has extensively studied the use of sustainable energy sources such as solar and wind power, and the industry continues to invest in their development. To enhance their reliability and achieve a better balance between supply and demand, these sources require tailored management approaches and storage systems that take into account their geographical specificities, and most importantly, their intermittent operation. Therefore, integrating these storage systems is a crucial topic in the design of HRESs [2].

An HRES is an electrical system comprising more than one energy source, at least one of which is renewable (such as wind, solar, hydro, tidal, geothermal, etc.). Depending on how they are interconnected to the grid, two types of HRES are identified in the literature: stand-alone HRES and grid-connected HRES [3]. Generally, HRESs that are connected to the main power grid does not require storage because the difference between intermittent energy production and the

energy demand can be offset by the regular electricity grid. On the other hand, in stand-alone HRESs, storage devices are extremely significant because they are used to store extra energy, which can later be reintroduced to satisfy the demand when there is an energy shortage.

Recently, with the growing interest in hydrogen technology, which is known for its advantages in terms of the long life of hydrogen tanks and lower costs compared to batteries, HRESs with hydrogen systems have been addressed in several studies [3]. These systems use batteries for short-term energy storage, while the fuel cell is operated for long-term storage. Hybridizing short- and long-term storage through the use of batteries and hydrogen-related technologies is a viable and attractive solution for stand-alone HRESs. In these configurations, the batteries are used as a buffer tank connected to a DC link or bus. The short-term temporal decoupling between production and demand is directly absorbed by the batteries, while in the long term, hydrogen is used as an energy carrier for the transfer between long periods of overproduction and phases of underproduction. The most commonly used electricity-hydrogen-electricity transformation chain is based on electrolyzer and fuel cell technologies. In this configuration, the surplus produced by renewable sources is employed to produce hydrogen via the electrolysis of water, which is then stored in a tank. The fuel cell is then used to generate electricity again from the chemical reaction between the stored hydrogen and oxygen, which is normally extracted from the ambient air [5].

Since several components are integrated into HRESs, the power of these systems must be properly designed to increase the overall energy efficiency of the HRES. Therefore, designing HRESs is a complicated task. It is essential to optimize the sizing of the HRES components to achieve economic, technical, and design objectives [6].

The sizing of the hybrid system components is a key step in designing a reliable and cost-effective system with a high penetration of renewable energy in the energy mix [7]. The objective is to find the optimal configuration that reduces the associated capital and maintenance costs, taking into account the constraints related to technical requirements, available energy, and load needs. This design problem requires a difficult solution involving nonlinear electrification optimization. Traditional approaches are unable to provide

optimal solutions due to the intricate nature of these discrete and continuous optimization problems [8]. As a result, meta-heuristic algorithms have been devised, taking inspiration from natural processes, to obtain the best possible solutions due to their extraordinary capability to achieve the desired result. Usually, these algorithms contain two major processes: exploration and exploitation. During the exploration stage, the algorithm is able to explore the global space, while the exploitation stage allows it to identify the optimal solution among all promising areas [9].

This research paper has made the following primary contributions:

- Employing a novel approach known as the Dandelion Optimizer (DO) [10] to establish the optimal size of an HRES in an isolated area in Lagouira City, Southwest Morocco.
- The proposed microgrid components are composed of PV, WT, Inverter, FC, HT, and EL units.
- The primary objective of this study is to minimize the cost of energy (COE), reduce the net present cost (NPC) and satisfy the reliability constraint, which is the loss of power supply probability (LSPS).
- The capabilities and effectiveness of the DO approach are evaluated by comparing it with widely known optimization algorithms such as GA and PSO.

This paper is structured in the following order: Section II briefly explains the proposed HRES; Section III provides an economic analysis of the system, describes the optimization problem, and details the energy management approach; Section IV discusses the simulation results; and finally, Section V outlines the study's conclusions.

II. DESCRIPTION OF THE PROPOSED HRES

The hybrid system studied in this paper consists of PV panels, Wind Turbines (WTs), inverters, Electrolyzer units (EL), Fuel Cell (FC) components, and Hydrogen Tank units (HT). Figure 1 presents the schematic configuration of the proposed HRES. The sizing of the various elements of hybrid energy systems has been designed based on the environmental conditions and manufacturer characteristics of the individual components, as described below.

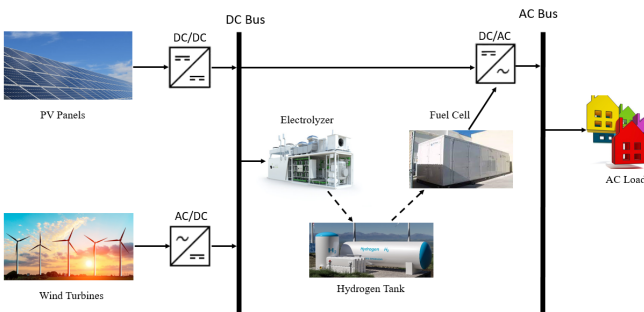


Fig. 1: Schematic diagram for hybrid energy system

A. PV generator model

A variety of models have been proposed in the literature to determine the output power of photovoltaic modules (P_{pv-out}). In this study, we use a simplified model that takes into consideration the solar radiation and ambient temperature, as shown in the equation below [11].

$$P_{pv-out}(t) = P_{pv-rated} \times \frac{G_t(t)}{1000} [1 + K_t((T_{amb} + (0.0256 \times G_t)) - T_{C,STC})] \quad (1)$$

The generated power of a photovoltaic (PV) module, denoted as P_{pv-out} , is measured in watts (W). The solar irradiance, represented by G_t , is measured in watts per square meter (W/m^2). The rated power of the PV module, denoted as $P_{pv-rated}$, refers to the power output at standard test conditions (STC) and is also expressed in watts. The temperature coefficient, K_t , is defined as $(-3.7 \times 10^{-3} (1/^\circ C))$. The cell temperature at STC, denoted as $T_{C,STC}$, is measured in degrees Celsius ($^\circ C$), while the ambient temperature, T_{amb} , is also measured in $^\circ C$.

B. Wind turbine model (WT)

The output power of a WT, denoted as $P_{WT}(t)$, can be determined using the following equation [12]:

$$P_{WT} = \begin{cases} 0 & \text{if } V(t) \leq V_{ci} \text{ or } v \leq V_{co} \\ P_{WT-rated} \times \frac{V(t) - V_{ci}}{V_r - V_{ci}} & \text{if } V_{ci} \leq v \leq V_{rat} \\ P_{WT-rated} & \text{if } V_{rat} \leq V(t) \leq V_{co} \end{cases} \quad (2)$$

where V_{ci} represents the cut-in speed, V_{rat} signifies the rated wind speed, V_f represents the furling speed, and $P_{WT-rated}$ represents the rating of a single wind turbine. The wind speed at the hub height, expressed as $V(t)$, is specific to the site and geographical location and may differ from the reference height. This can be expressed using the following formula [13]:

$$V(t) = V_r(t) \left(\frac{H_{WT}}{H_r} \right)^\gamma \quad (3)$$

In this equation, the wind speed at the height of the wind turbine (H_{WT}) is represented by $V(t)$, while the wind speed at the reference height (H_r) is denoted by $V_r(t)$. The friction coefficient, which is typically 1/7 for low-roughness surfaces and well-exposed sites, is denoted by the Greek letter gamma (γ).

C. Electrolyzer model (EL)

The Electrolyzer generates oxygen and hydrogen through the process of water electrolysis. When the hybrid system experiences a power shortage, the hydrogen stored in the hydrogen storage tank is used to meet the load demand by being transferred to the Fuel Cell (FC). The model of the Electrolyzer can be described as follows [14]:

$$P_{el_ht} = P_{ren_el} \times \eta_{el} \quad (4)$$

where the output power of the electrolyzer is denoted as $P_{el_{ht}}$ and is determined by the amount of renewable energy input, which is expressed as $P_{ren_{el}}$. The final power output is also influenced by the efficiency of the electrolyzer, represented by η_{el} .

D. Hydrogen storage model (HT)

The formula for calculating the energy of the hydrogen storage tank is expressed as follows [15]:

$$E_{ht}(t) = E_{ht}(t-1) + P_{ren_{el}}(t) \times \Delta t - P_{ht_{fc}}(t) \times \Delta t \times \eta_{ht} \quad (5)$$

where the energy stored in the tank at two different times, namely time t and $t-1$, are denoted as $E_{ht}(t)$ and $E_{ht}(t-1)$, respectively. The variable $P_{ht_{fc}}$ represents the total power that is supplied to the FC from the HT at time t , while η_{ht} indicates the efficiency of energy storage in the HT. The time interval, which has a fixed value of 1 hour in this case, is indicated by Δt .

$$M_{ht}(t) = \frac{E_{ht}(t)}{HHV_{H_2}} \quad (6)$$

In addition, the variable M_{ht} represents the quantity of hydrogen that is fed into the HT storage, and HHV_{H_2} represents the energy content of hydrogen, which is generally known to be 39.72 kWh/kg.

E. Fuel cell model (FC)

The power generated by the final output FC is based on the amount of hydrogen received, thus, its model is expressed as follows [15]:

$$P_{fc_{inv}} = P_{ht_{fc}} \times \eta_{fc} \quad (7)$$

where the efficiency of the FC referred to η_{fc} .

F. Inverter

To distribute all of the generated power by renewable energy sources and stored in energy storage devices as AC loads for consumers, an inverter must be employed. The inverter is represented by the following model:

$$P_{inv_{load}} = (P_{ren_{inv}} + P_{fc_{inv}}) \times \eta_{inv} \quad (8)$$

where the energy input to the inverter from the fuel cell is represented by $P_{fc_{inv}}$, while $P_{ren_{inv}}$ denotes the energy input to the inverter from the renewable source.

III. PROBLEM FORMULATION

A. Economic and cost analysis

The objective of this research is to minimize the cost of energy (COE) produced by the HRES proposed while ensuring a reliable power supply to consumers. The COE is measured in dollars per kilowatt-hour (kWh), and is calculated as follows [16]:

$$COE(\$/kWh) = \frac{NPC(\$)}{\sum_{t=1}^{t=8760} P_{load}(kWh)} \times CRF \quad (9)$$

The Net Present Cost (NPC) represents the overall expenses incurred during the lifetime of the project, including capital costs (C), replacement costs (R), and operation and maintenance costs (OM). It can be represented as follows:

$$NPC = C + OM + R \quad (10)$$

The Capital Recovery Factor (CRF) can be computed using the following equation:

$$CRF(i, y) = \frac{i(1+i)^y}{(1+i)^y - 1} \quad (11)$$

where in this study, the interest rate (i) was fixed at 6% per year and the lifespan of the system (y) was determined to be 20 years.

B. Constraints

To determine the optimal dimensions of the proposed hybrid system, an optimization algorithm called DO has been utilized. The optimization process is subject to the following constraints:

- The minimum and maximum size of hybrid system components.

$$N_{i-min} \leq N_i \leq N_{i-max} \quad (12)$$

$$SOC_{tank-min} \leq SOC_{tank} \leq SOC_{tank-max} \quad (13)$$

where the system's lower and upper limits for component numbers or capacities, such as the number of photovoltaic and wind units, inverter power transmitted to the load (kW), electrolytic power (kW), fuel cell power (kW), and hydrogen tank mass (kg), are indicated by N_{i-min} and N_{i-max} . Moreover, the minimum and maximum levels of hydrogen tank energy state of charge (kWh), designated as $SOC_{tank-min}$ and $SOC_{tank-max}$, are also considered.

- To assess the technical design of the hybrid system, the reliability constraint is determined based on the maximum LPSP ($LPSP_{max}$).

$$LPSP \leq LPSP_{max} \quad (14)$$

where the $LPSP_{max}$ is considered as 0.05 (5%).

C. Energy management strategy

The output capacity of PV (photovoltaic) and WT (wind turbine) renewable energy sources is compared to the energy demand at each hour (h). The functioning combined PV, WT, and FC energy systems operate in the following manner.

- When the total generated power from both PV panels and WTs (P_{ren}) meets the demand for electricity. (P_{load}):

$$P_{ren} = \frac{P_{load}}{\eta_{inv}} \quad (15)$$

TABLE I: Economic details of the hybrid system components [13], [21].

Component	CC	RC	OM	Rated Size	Lifetime (years)	Availability (%)
PV	2000	1800	20	1 kW	20	100
WT	3200	3000	32	1 kW	20	100
Inverter	800	750	7	1 kW	15	100
Electrolyser	2000	1500	20	1 kW	20	100
Fuel cell	3000	2500	40	1 kW	5	100
Hydrogen tank	1300	1200	15	1 Kg	20	100

In this scenario, the entire demand for power is met and the LPSP is equal to zero. η_{inv} represents the efficiency of the inverter.

- The generated power by PV panels and WTs exceeds the load demand P_{load} :

$$P_{ren} \geq \frac{P_{load}}{\eta_{inv}} \quad (16)$$

The surplus energy is directed into the electrolyzer, leading to the creation of hydrogen which is then stored in a pressurized hydrogen tank as reserve energy. As a result, LPSP equals zero as the entire demand is being met.

- When the power produced by photovoltaics (PVs) and wind turbines (WTs) does not match the total power demand, meaning that the output of the renewable sources is less than P_{load} :

$$P_{ren} < \frac{P_{load}}{\eta_{inv}} \quad (17)$$

To make up for the shortfall in energy demand, the process entails transferring hydrogen to the fuel cell, producing electricity, and distributing it to fulfill the energy demand. In cases where the combined output from photovoltaics, wind turbines, and the fuel cell falls short of the energy demand, a portion of the demand will be cut off. As a result, the level of reliability (LPSP) is expressed as a value between 0 and 1.



Fig. 2: Dandelions floating¹.

¹<https://pixabay.com/photos/dandelion-heaven-flower-nature-463928/>

TABLE II: Technical details of the hybrid system components [13], [21].

PV Panel	
Nominal temperature	25°C
Temperature under nominal operation	45°C
Temperature coefficient	-3.7×10^{-3}
Instantaneous radiation under STC	1000 W/m ²
Wind turbine	
Rated wind speed	11 m/s
Height of turbine	20 m
Cut-in speed (m/s)	3 m/s
Cut-out speed (m/s)	25 m/s
Hydrogen tank	
HHV_{H_2}	39.72 kWh/Kg
Electrolyzer	
Efficiency (η_{EL})	75%
Fuel cell	
Efficiency (η_{FC})	60%
Inverter	
Efficiency (η_{Inv})	95%

D. Dandelion optimizer (DO)

The DO is a novel nature-inspired optimization algorithm proposed by Shijie Zhao et al. in 2022 [10]. DO simulates the process of dandelion seeds dispersing from their parent plant, with each seed representing a possible solution to an optimization problem (see Figure 2). Dandelions exhibit diverse flight patterns due to the influence of weather and wind, which lead to their dispersal across various locations. The algorithm operates based on three stages, where seeds spiral upwards to reach a specific height during the initial rising stage. In the floating stage, seeds drift with the wind, covering a broad area. Finally, in the falling stage, seeds drop to the ground and distribute their offspring. By using these three stages, DO searches for optimal solutions in an efficient manner.

IV. RESULTS AND DISCUSSIONS

To evaluate the effectiveness of the proposed optimization method, we conducted a case study involving the design of an HRES in Lagouira, a city located in the southern region of Morocco (latitude: 20.8333, longitude: -17.0917). The study analyzed hourly average solar radiation on a horizontal plane and temperature data over a year, as shown in Figure 3 and Figure 4, respectively. Additionally, Figure 5 illustrates the typical hourly wind speed in the region during the same time-frame, along with the wind direction at a height of 10 meters above the ground. We obtained wind speed, solar radiation, and temperature data from the NASA Surface Meteorology

TABLE III: The results of hybrid system designing for LPSP=5%

Parameter Algorithm	N_{PV}	N_{WT}	P_{EL} (kW)	P_{FC} (kW)	M_{HST} (Kg)	P_{Inv} (kW)	LPSP	NPC (\$)	CEO (\$/kWh)
DO	60	55	18.85	20.22	74.93	49.46	0.0499	825,395	0.2674
GA	59	55	19.29	20.33	74.75	49.96	0.0499	825,395	0.2674
PSO	60	54	18.87	19.73	75.03	49.66	0.0496	825,395	0.2674

and Solar Energy website for a period of 20 years in the selected location². Figure 6 shows the IEEE load demand profile throughout the year. The microgrid project in the study was planned to have a lifespan of 20 years and an interest rate of 6%. Table I provides the economic parameters for the microgrid components, while Table II contains the technical parameters.

A. Comparison with well-known techniques

In this research, a time interval of one hour was used for all input parameters during simulation and analysis. The hourly load curve was observed throughout the day. To guarantee the robustness and stability of the proposed algorithms and avoid randomness, optimization programs based on DO, GA, and PSO algorithms were implemented 20 times with 100 iterations and a maximum population size of 20, and the lowest objective function value was recorded. Figure 8 displays the convergence curves for the hybrid system using DO, GA, and PSO methods over the 20 runs. According to the results, the proposed DO algorithm consistently provides objective function values that fall within a narrow range. This demonstrates that DO is a stable method for addressing optimization problems.

From the results in Table III, it can be inferred that the DO, GA, and PSO methods provide a minimum COE of 0.2674 \$/kWh, which satisfies the predefined value of LPSP ($LPSP \leq LPSP_{max}$) with a value of 0.0499. The convergence curves of the best-obtained solution of DO, GA, and PSO algorithms at 100 iterations are shown in Figure 7. DO exhibits the best performance and requires less computing time to obtain the optimal solution due to its faster convergence process.

To accurately compare all the methods, statistical analyses were performed on the objective function values obtained from 20 individual runs for the studied case using both parametric and non-parametric measures. Parametric measures included minimum, maximum, and mean objective function values, while non-parametric measures included median, standard deviation (SD), mean absolute error (MAE), kurtosis, and skewness. When the skewness of the variation of COE is at its minimum, it indicates that the distribution of this statistical measure is very similar to the normal distribution curve. On the other hand, when the kurtosis is at its lowest point, it implies that the peak of the curve is relatively flat compared to other optimization techniques. The statistical results for the DO, GA, and PSO algorithms are presented in Table IV. Based

TABLE IV: Statistical characteristics of the COE results of different algorithms

Statistical Parameters	DO	GA	PSO
Best	0.2674	0.2674	0.2674
Worst	0.2738	0.3063	0.3412
Mean	0.2698	0.2764	0.2787
Median	0.2694	0.2738	0.2696
SD	1.90E-03	9.50E-03	2.03E-02
MAE	0.2698	0.2764	0.2787
Kurtosis	3.0514	5.8537	6.0672
Skewness	1.0523	1.666	2.0876

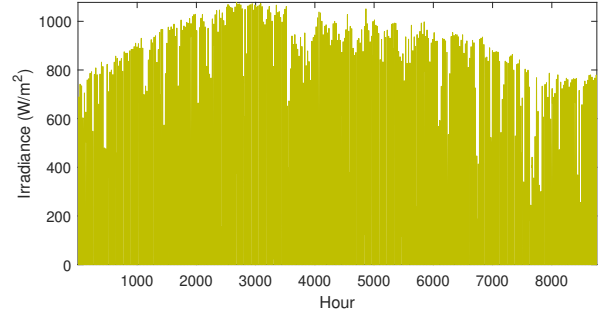


Fig. 3: Hourly solar insolation profile data during a year.

on the results, the DO algorithm outperforms the GA and PSO techniques in all metrics.

Figure 9 illustrates the power output of the hybrid PV/WT/FC system components with a low LPSP value of 5% throughout the year when employing DO. The figure displays the total power generated by the photovoltaic panel and wind turbine, as well as the power injected into the electrolyzer and the inverter by the hybrid system. Additionally, it indicates the energy stored in hydrogen and the LPSP variations over the year.

B. Comparison with previous studies in literature

Table V provides a comparative analysis of the COE per kWh reported in previous studies, which employed various methodologies and were conducted in different regions. It is worth noting that the cost of energy production is subject to variation, depending on the unique patterns of radiation and wind speed that characterize each geographic area. Consequently, the differences in costs across the studies do not necessarily indicate the superiority or inferiority of any particular method; rather, they reflect the potential of each region for implementing the proposed hybrid systems. This information can be particularly valuable for energy system designers, as it allows them to evaluate the economic feasibility of renewable

²<https://www.soda-pro.com/web-services/meteo-data/>

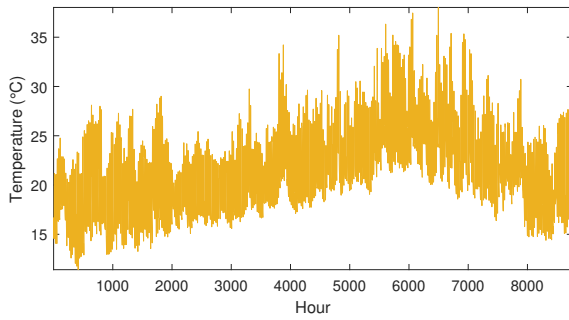


Fig. 4: Hourly temperature profile data during a year.

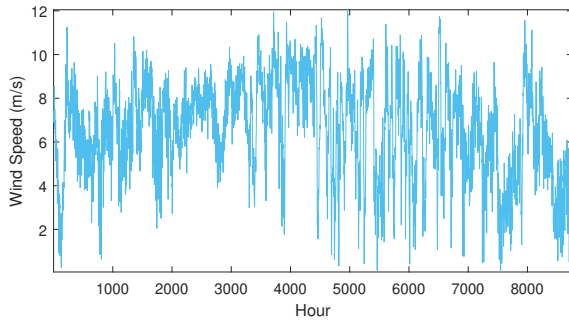


Fig. 5: Hourly wind speed data during a year.

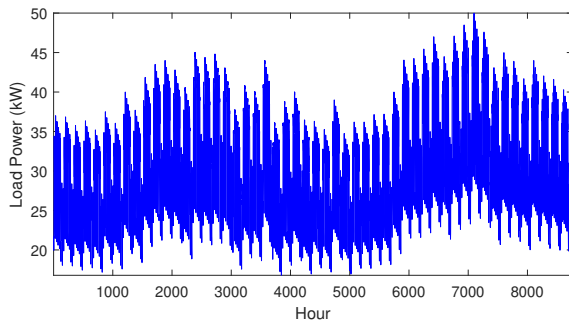


Fig. 6: IEEE load demand data during a year.

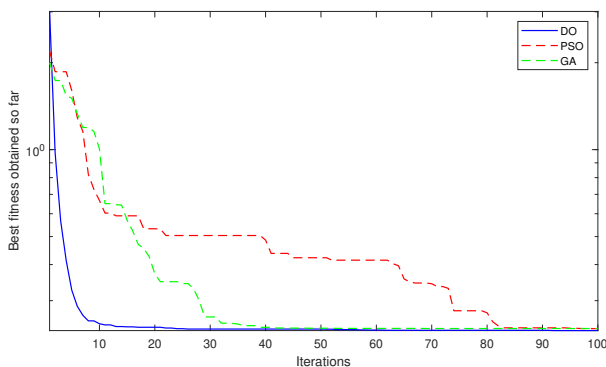


Fig. 7: Convergence curve of DO, GA, and PSO for the proposed hybrid system.

energy projects based on the available radiation and wind speed potential in a given area.

V. CONCLUSION

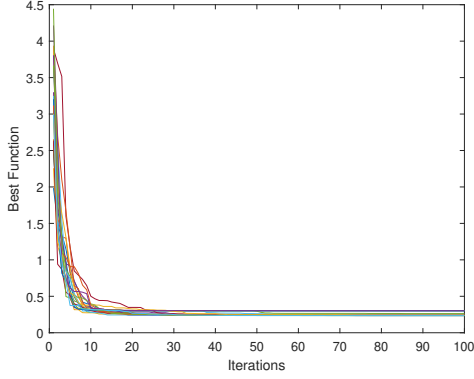
This paper proposes using the Dandelion Optimizer (DO) to address the optimization problem of designing an isolated hybrid microgrid in Lagouira City, Morocco. The microgrid consists of photovoltaic panels, wind turbines, hydrogen tank units, electrolyzer units, and a fuel cell system for a rural region. The main goal of this project is to minimize the cost of energy for the hybrid system while satisfying reliability constraints to fulfill the load demand and manage the power flow among the microgrid system's components. The study compares the results of the DO algorithm with other prominent optimization methods, including the Genetic Algorithm (GA) and Particle Swarm Optimization (PSO), to determine DO's superiority. Compared to GA and PSO, the DO algorithm demonstrates better convergence characteristics and requires less time to obtain an optimal solution. Therefore, DO is considered an excellent option for properly sizing a microgrid system. The extensive study conducted in this research has demonstrated the significant benefits of the suggested algorithm in addressing optimal design issues for hybrid renewable energy systems. These findings have important implications for researchers across various scientific disciplines. Therefore, this article is presented to facilitate knowledge sharing and to contribute to the advancement of research in this area.

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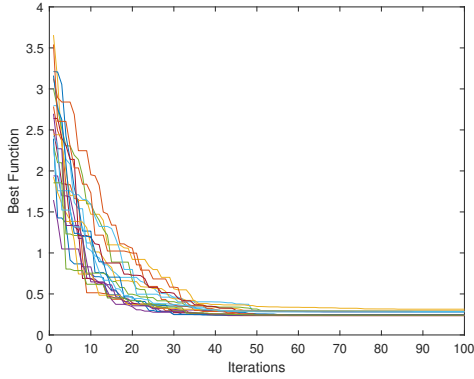
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TABLE V: COE of previous studies in literature

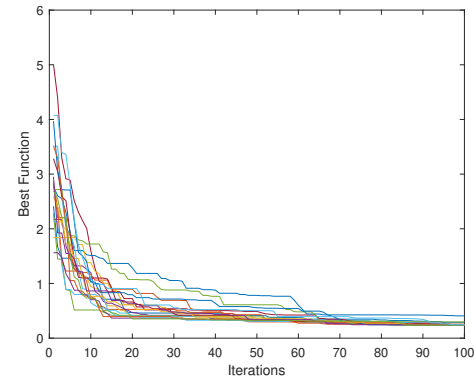
Ref	System	Method	Region	COE (\$/kWh)
[17]	PV/wind/DG/battery	Homer	Kudu village, Nigeria	0.259
[18]	PV/WT/FC	IAEO	Ataka, Egypt	0.412
[19]	PV/Wind/Diesel/Bat	IAOA	El Kjharga oasis, Egypt	0.2606
[20]	PV/WT/Tidal/FC	WOA	Yazd, Iran	0.4477
[21]	PV/FC/Battery	Homer	Jeddah, Saudi Arabia	1.40
Proposed	PV/WT/FC	DO	Lagouira, Morocco	0.2674



(a) DO

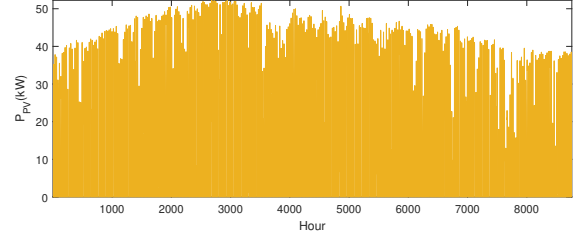


(b) GA

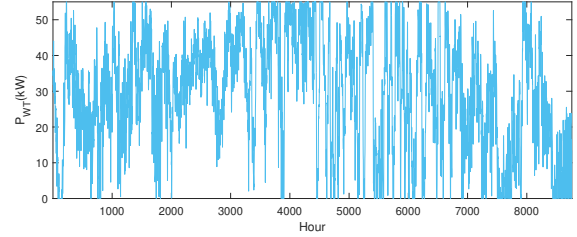


(c) PSO

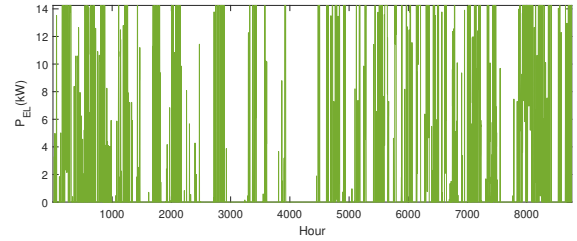
Fig. 8: Convergence curves over 20 runs of the proposed algorithms



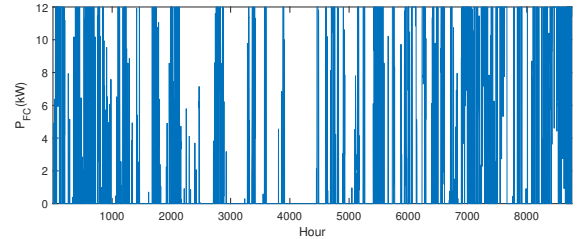
(a) Generated power with PV panels



(b) Generated power with WTs



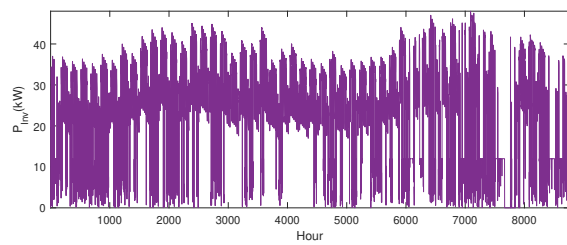
(c) Electrolyzer power



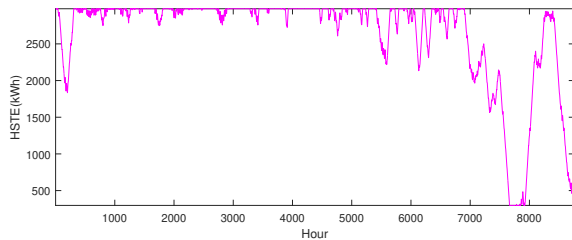
(d) Fuel cell power

Fig. 9: Power contribution of the proposed system components

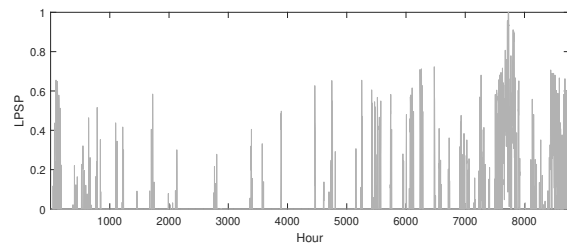
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(e) Injected power from the hybrid system to the inverter



(f) Hydrogen tank energy



(g) Variations of LPSP during a year

Fig. 9: (Continued)

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